

3 A program is written to perform different individual processes using arrays.

(a) A 1D array stores 10 integers.

(i) A recursive function `RecursiveCount()` takes three parameters:

- `ArrayCopy`, an array of integers
- `NumberElements`, the number of elements in the array of integers
- `DataToFind`, an integer data to find in the array.

The function counts and returns the number of times `DataToFind` is in `ArrayCopy`

The recursive algorithm:

- returns 0 if there are no elements in `ArrayCopy`
- compares the first element in `ArrayCopy` with `DataToFind`
 - if the element matches, the function returns 1 added to the return value from a recursive call (passing the array without the first element)
 - if the element does **not** match, the function returns the return value from a recursive call (passing the array without the first element).

Write program code for `RecursiveCount()`

Save your program as **Question3_N25**.

Copy and paste the program code into part **3(a)(i)** in the evidence document.

[6]

(ii) The main program stores the following data in a 1D array of integers in the order given:

0 5 1 2 5 9 9 6 5 0

The main program calls `RecursiveCount()` with the parameters:

- 0 as the data to find
- 10 as the number of elements
- the array of 10 integers.

The return value is output.

Write program code for the main program.

Save your program.

Copy and paste the program code into part **3(a)(ii)** in the evidence document.

[3]